Pac-Man Lab 4

Reconstructed learning dialogue

User

I need help understanding my Lab 4 assignment. Please guide me with hints and checks so I can work through the code and try the game manually.

Claude

The reference describes one deliberate bug and three empty hooks: ghost_color, on_pellet_eaten, and bonus_life_threshold. Start by reading the handout and README, saving the starter version, and running the game to understand its current behaviour.

User

I will keep an unchanged copy before editing. How should I investigate why eating a power pellet does not start frightened mode?

Claude

Trace the pellet symbol from the maze definition to eat(). The screenshot identifies a case mismatch: eat() checks uppercase "O", but the maze uses lowercase "o". Inspect those locations yourself before changing the condition.

User

I will compare the symbols and make that small correction myself. What should I check when I run the game again?

Claude

Eat a power pellet and observe whether frightened mode starts and ghosts reverse direction. The reference reports fright_left becoming 3.0. Verify that value in your own run and also check that ordinary pellets still behave correctly.

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User

For ghost_color, I want to choose and implement the colours myself. Can you explain what the hook should return?

Claude

According to the reference, return a distinct frightened tint for each ghost while frightened mode is active, and None otherwise. Inspect the caller to see how None preserves the normal rendering. Choose colours that remain distinguishable in the maze.

User

I will try my colour choices in the game and check the normal appearance after frightened mode ends. How should I approach on_pellet_eaten?

Claude

The described feature spawns a bonus fruit worth 100 points when 30 pellets remain and flashes the HUD on the last pellet. First locate the remaining-pellet count, fruit state, scoring code, and reset(). Then implement and test one behaviour at a time.

User

I will inspect whether the hook runs before or after the pellet count changes. I will also check that fruit is not repeatedly spawned and that restarting clears the state.

Claude

Those checks will help catch off-by-one and stale-state errors. Test the transition around 30 remaining pellets, fruit collection and the 100-point award, then the last pellet and a restart. Record what actually happens rather than assuming the feature works.

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User

For bonus_life_threshold, why does the reference suggest 1000 instead of 10000? I want to understand the choice before implementing it.

Claude

The screenshot states that the maze has 130 pellets and uses 1000 as a reachable threshold. Verify the scoring rules in your version before relying on that reasoning. Test reaching or crossing the threshold and check that the same bonus is not awarded repeatedly.

User

I will do the changes and manual checks in my local game. What evidence should I collect for the assignment?

Claude

Keep the unchanged starter code and record real game-window footage before and after your edits. Include visible demonstrations of frightened mode, distinct ghost colours, the extra life, bonus fruit, and the win screen where required. Simulated clips should not be presented as real gameplay.

User

I will review the videos and organise my files myself. How should I prepare the final repository submission?

Claude

The reference repository uses SELABS/LabN folders, filenames ending in _Affan_Dumba_PES1UG24CS280, and a README table per lab. Check the actual handout and repository, follow that structure for Lab 4, and review your diff before pushing to your own submission repository. The reference says not to raise a PR to the original repository. Include an accurate AI-use record.